Assignment 6

2(a)(i). Archived summit tweets

Tweet archive checks; repeated text is retained

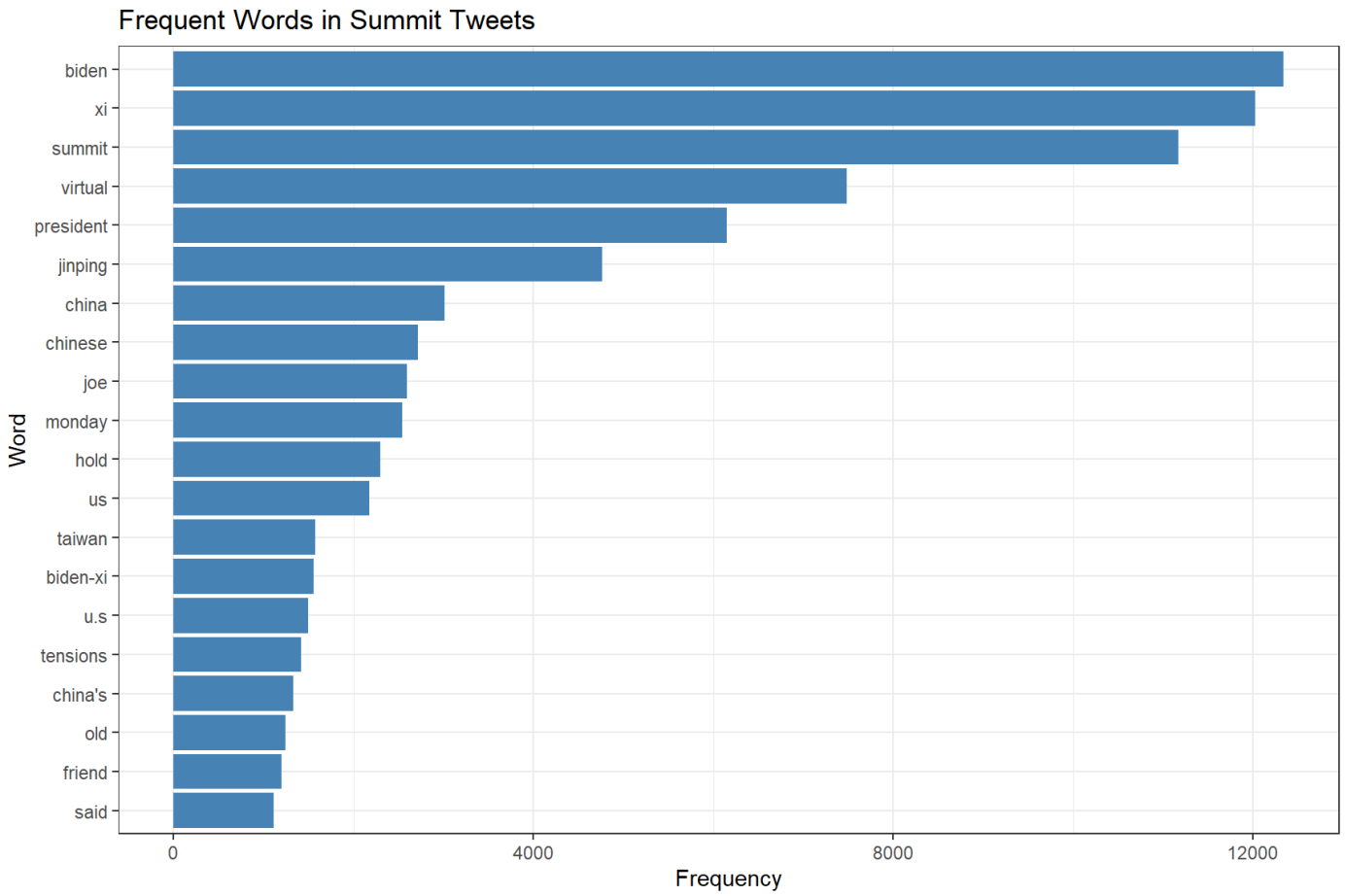
archived_rows	missing_or_blank	retained_rows	repeated_text
14520	0	14520	9059

Frequent words

Most frequent content words

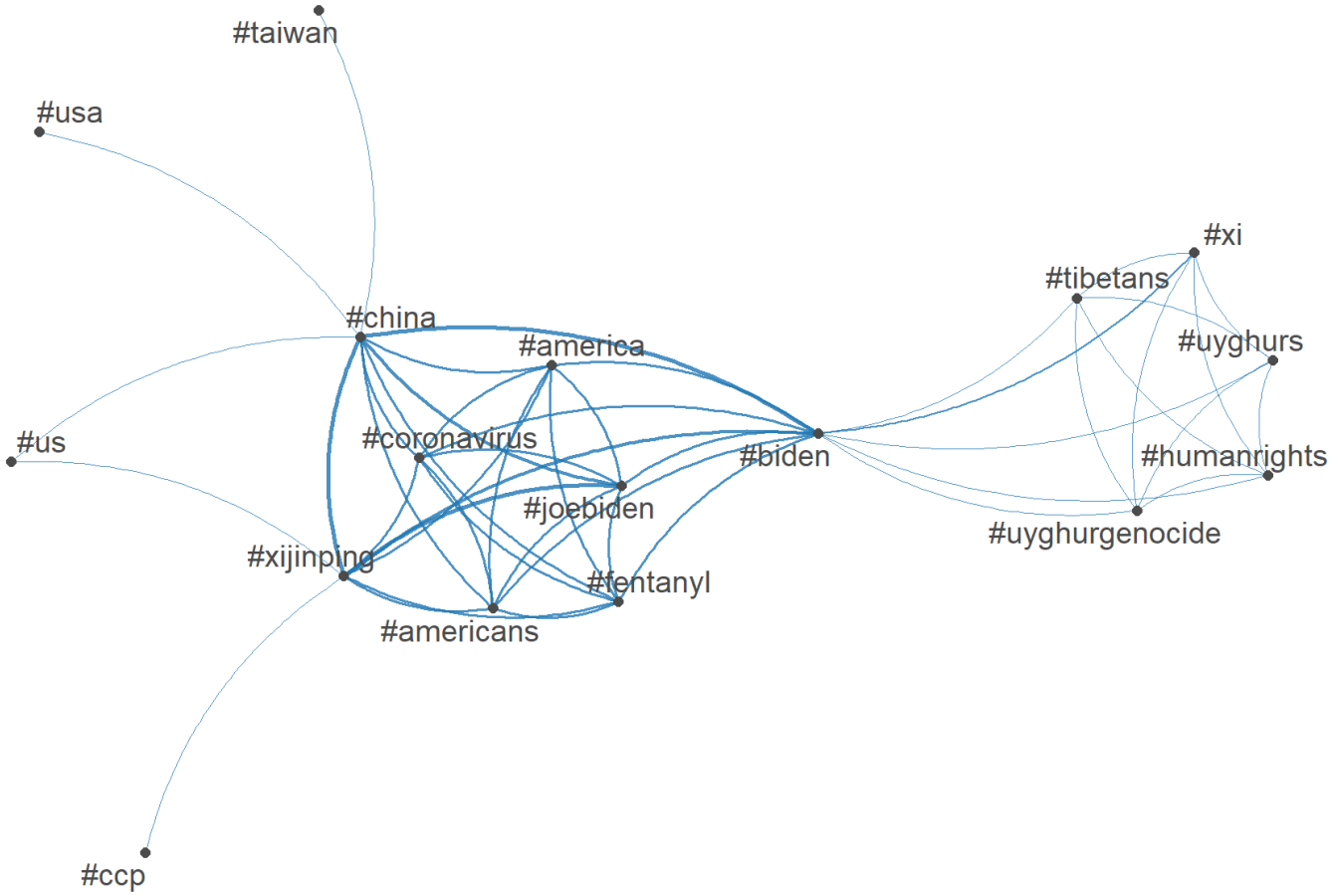
feature	frequency	rank	docfreq	group
biden	12338	1	9938	all
xi	12027	2	10472	all
summit	11167	3	11047	all
virtual	7483	4	7408	all
president	6148	5	4256	all
jinping	4766	6	4722	all
china	3011	7	2847	all
chinese	2716	8	2690	all
joe	2598	9	2578	all
monday	2540	10	2506	all
hold	2301	11	2291	all
us	2179	12	2046	all
taiwan	1578	13	1506	all
biden-xi	1556	14	1545	all
u.s	1493	15	1398	all
tensions	1420	16	1412	all
china’s	1330	17	1313	all
old	1246	18	1208	all
friend	1196	19	1181	all

feature	frequency	rank	docfreq	group
said	1113	20	852	all



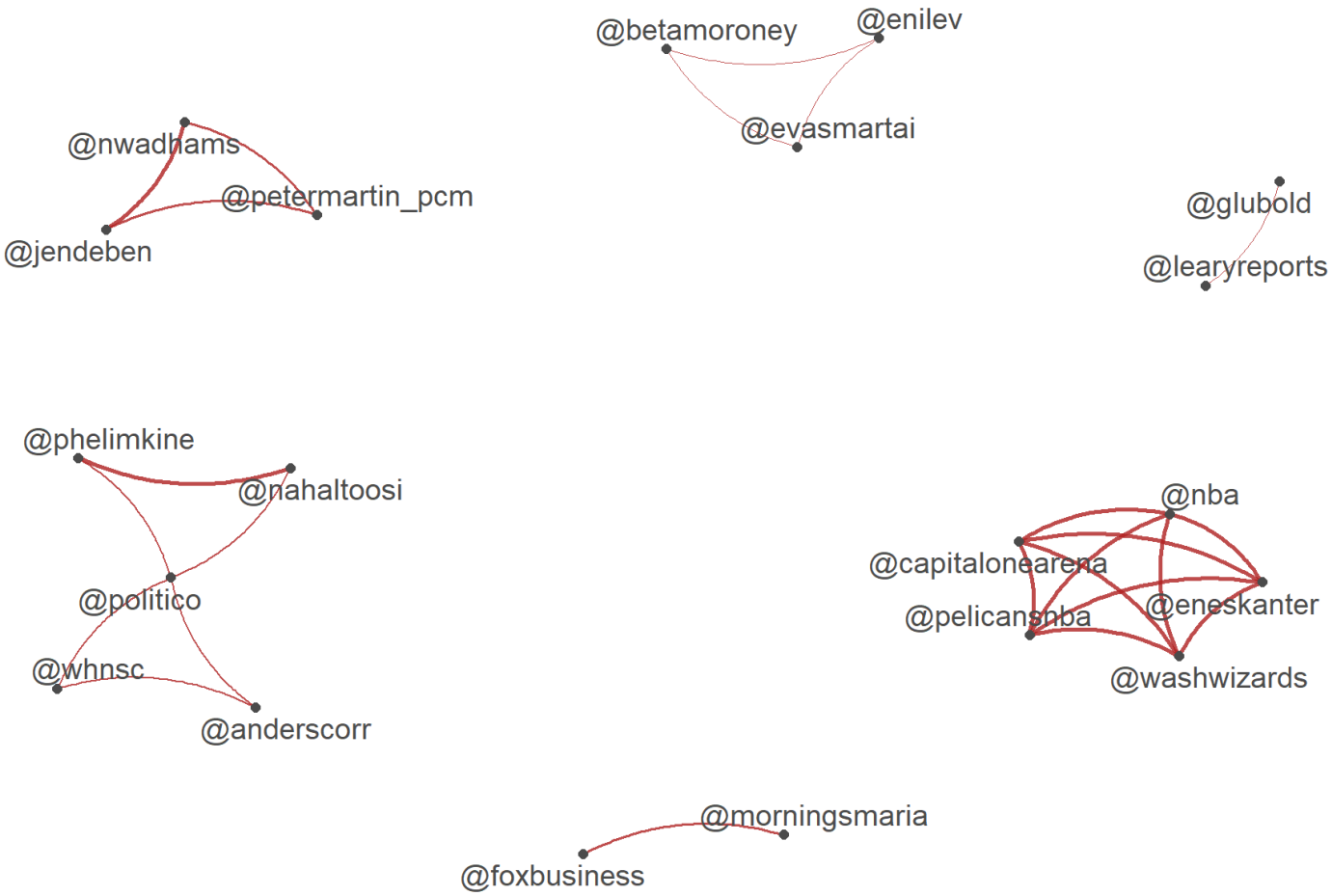
Hashtag network

```
[1] "#china"       "#biden"       "#xijinping"   "#joebiden"    "#america"
[6] "#americans"   "#coronavirus" "#fentanyl"    "#xi"          "#us"
```



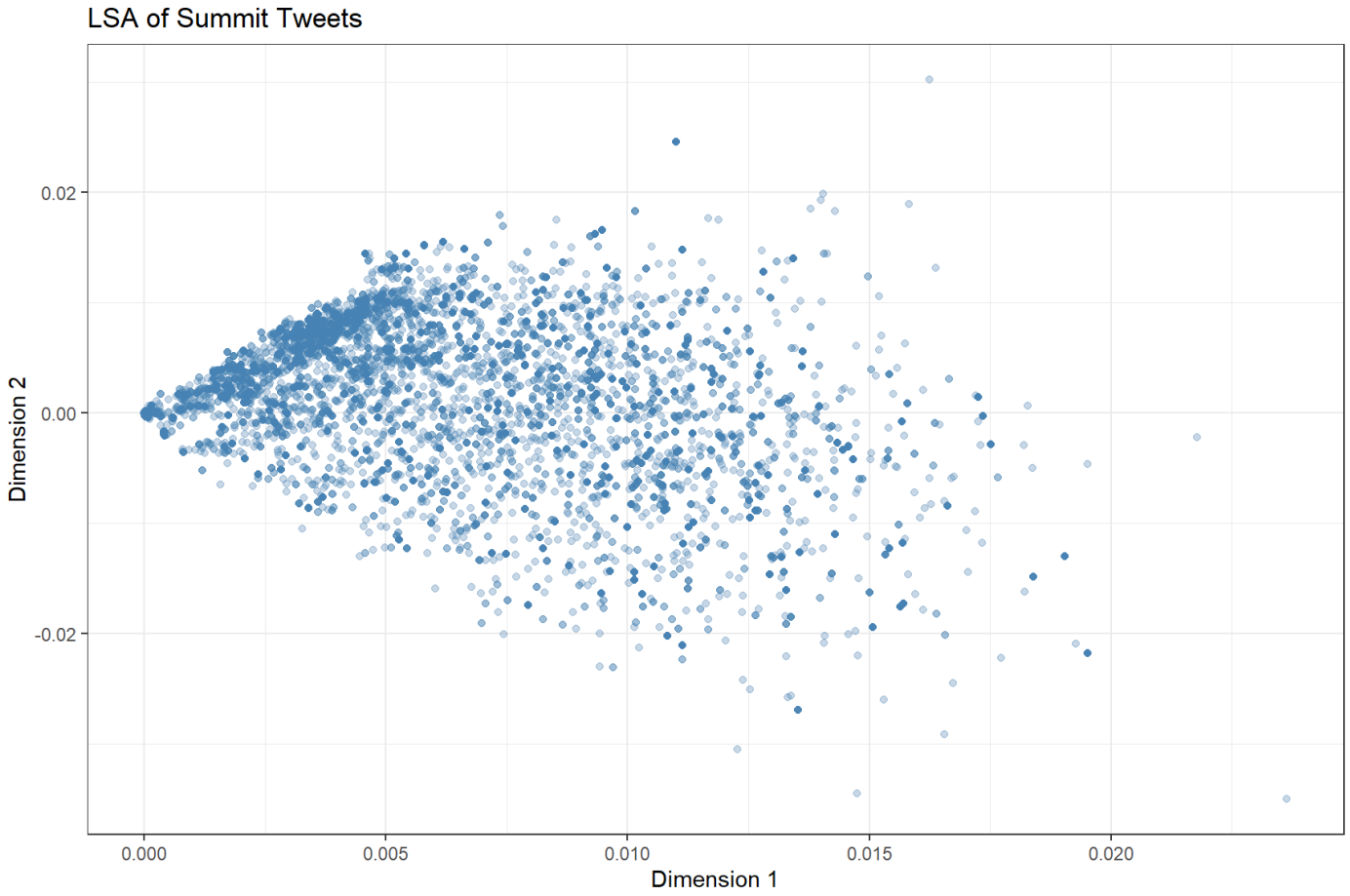
Mention network

```
 [1] "@potus"           "@politico"        "@joebiden"        "@jendeben"
 [5] "@eneskanter"      "@nwadhams"        "@phelimkine"      "@nahaltoosi"
 [9] "@nba"             "@washwizards"     "@pelicansnba"     "@capitalonearena"
[13] "@kevinliptakcnn"  "@foxbusiness"     "@morningsmaria"   "@scmpnews"
[17] "@petermartin_pcm" "@nytimes"         "@uyghur_american" "@kaylatausche"
```



Latent Semantic Analysis

```
[1] 14520     4
```




```
            Dimension1
text5077  -4.294725e-19
text12021 -4.083143e-19
text7221  -3.958900e-19
text1935  -3.957985e-19
text12163 -3.890606e-19

tweet
text5077  มิวายงานอ่า ปะหานาหิบ่ดิใจ ใบด้น ดะละปะหานปะเทดจิน ท่านสิ จิ้นผิง จะปะชุมสุดยอด ทางออนไลน์ ในอาทิดหน้า
https://t.co/NTiNbYVHWs
text12021
https://t.co/S5q3q1TLTI
text7221
https://t.co/aYLpjnhUVy
text1935
https://t.co/fjB0UfMGW7
text12163
https://t.co/Dmlc5xdFxc

         Dimension1
text9606 0.02363972
text6997 0.02178694
text9017 0.01952424
text5481 0.01952411
text5765 0.01952411
```

tweet
text9606   @GOP ...annual climate summit, COP26. "the communist leader." "The communist leader?" That's your guys' name for Xi? "accountable for the spread of COVID?" What a ridiculous "question." Biden wasn't president last year, when the pandemic was raging across the globe. He didn't...
text6997 US-China virtual summit..\n'Old friends'? For Xi Jinping and Joe Biden, not necessarily ..\nBiden urges for 'guardrails' against conflict.. \nPresident Joe Biden and Xi Jinping met nearly four hours. Biden is trying to engage in what his administration calls .. https://t.co/TDkpMLnaVH
text9017   Reports: Biden & Xi to Hold Virtual Summit Next Week. \nXi shall claim the the POWER OUTAGE to cut off the video link.\nXi will not bail Joe out of recession, inflation, accept worthless USD for trade, buying US debts, etc.\nJoe will station troops in Taiwan, FONOP in SCS, etc.
text5481   China’s readout of Xi’s summit with Biden. Thread:\n\nTaiwan’s a key topic. In the call, Xi blames Taipei for relying on US to seek independence. “We have patience,” he said, “but if Taiwan independence forces... breach the red line, we have no choice but take drastic measures.” /1
text5765   China’s readout of Xi’s summit with Biden. Thread:\n\nTaiwan’s a key topic. In the call, Xi blames Taipei for relying on US to seek independence. “We have patience,” he said, “but if Taiwan independence forces... breach the red line, we have no choice but take drastic measures.” /1

2(a)(ii). GovInfo presidential documents

Saved CPD corpus by administration

administration	documents
Biden	183
Trump	169

Xi Jinping in context

Keyword-in-context with 20 matches.

```
   [DCPD-202100229, 3342:3343]
   [DCPD-202100229, 3478:3479]
   [DCPD-202100238, 1722:1723]
   [DCPD-202100238, 2340:2341]
   [DCPD-202100238, 2428:2429]
   [DCPD-202100258, 7752:7753]
   [DCPD-202100258, 7777:7778]
 [DCPD-202100258, 11169:11170]
       [DCPD-202100259, 50:51]
     [DCPD-202100259, 357:358]
   [DCPD-202100271, 2991:2992]
   [DCPD-202100271, 5106:5107]
```

[DCPD-202100295, 3253:3254]
[DCPD-202100295, 3306:3307]
[DCPD-202100347, 9319:9320]
[DCPD-202100421, 4600:4601]
[DCPD-202100447, 3788:3789]
[DCPD-202100453, 1709:1710]
[DCPD-202100453, 1804:1805]
[DCPD-202100500, 3406:3407]

and former Prime Minister Enda Kenny of Ireland; former President Barack Obama; President
; O'Neill, Michelle; Psaki, Jennifer R.; Sloat, Amanda;
about it. It's about who we are. Closing comment: I was with
, he referred to Mayor Michael E. Duggan of Detroit, MI; President
; Obama, Barack; Walensky, Rochelle P.; Walke, Henry;
touch a smidgen of what the relationship with China really is about. I've known
the time I left office as Vice President, I had spent more time with
Obama, Barack; Putin, Vladimir Vladimirovich; Trump, Donald J.;
— Q. Do you have a comment on the Georgia election law? President
Washington, DC. Names: Johnson, Boris; Putin, Vladimir Vladimirovich;
ones. We just have to imagine again. I had a long discussion with
. Mitchell; Peduto, William M.; Wolf, Thomas W.;
Anthony S. Fauci; Senate Minority Leader A. Mitchell McConnell; and President
, Anthony S.; Francis, Pope; McConnell, A. Mitchell;
Jr.; Schumer, Charles E.; Trump, Donald J.;
.; Tidd, Mark L.; Van Ovost, Jacqueline D.;
G.; Ross, Clinton A.; Scott, Robert C.;
ARNG (Ret.), former adjutant general, Delaware National Guard; President
Jennings, Kathleen; Putin, Vladimir Vladimirovich; Vavala, Francis D.;
see just straightforward dealing with China. And again, we're not looking—as I've told

| Xi Jinping |
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| Xi Jinping |

of China; and White House Press Secretary Jennifer R. Psaki. He also
. Subjects: China: President; Civil rights: Minorities:: Minority rights
in China. I spent more time with him, I'm told, than any
of China, in his former capacity as Vice President of China; former President
. Subjects: China: Prime Minister; Diseases: Coronavirus, domestic prevention efforts
for a long time. Allegedly, by the time I left office as Vice
than any world leader had, because President Obama and the Chinese President Hu decided
. Subjects: Afghanistan: President; Afghanistan: U.S. military forces::
of China/ President Vladimir Vladimirovich Putin of Russia/ Prime Minister Boris Johnson of
. Subjects: China: President; Communications: News media:: Presidential interviews
, the leader of China, when he called to congratulate me. We spent
. Subjects: Business and industry: Domestic investment, promotion efforts; China:
of China. Categories: Addresses and Remarks: COVID–19 vaccination efforts; Interviews With
; Yellen, Janet L. Subjects: China: President; Congress: Senate
. Subjects: Afghanistan: U.S. military forces:: Deployment; Arms and
. Subjects: Armed Forces, U.S.: Military families; Armed Forces,
. Subjects: Afghanistan: U.S. military forces:: Deployment; Armed Forces
of China; and President Vladimir Vladimirovich Putin of Russia. He also referred to
. Subjects: Armed Forces, U.S: Military families; Armed Forces, U.S
myself, I'm not looking for conflict. Where we can cooperate, we'll cooperate

Keyword-in-context with 20 matches.

[DCPD-202100229, 2790]
[DCPD-202100229, 3342]
[DCPD-202100229, 3478]
[DCPD-202100238, 1722]
[DCPD-202100238, 2340]
[DCPD-202100238, 2428]
[DCPD-202100258, 7752]
[DCPD-202100258, 7777]
[DCPD-202100258, 8562]
[DCPD-202100258, 11169]
[DCPD-202100259, 50]
[DCPD-202100259, 76]
[DCPD-202100259, 357]
[DCPD-202100271, 2991]
[DCPD-202100271, 5106]
[DCPD-202100295, 3149]
[DCPD-202100295, 3197]
[DCPD-202100295, 3253]
[DCPD-202100295, 3306]
[DCPD-202100347, 3361]

when they ask me," Can I define America?" — like President
and former Prime Minister Enda Kenny of Ireland; former President Barack Obama; President
; O'Neill, Michelle; Psaki, Jennifer R.; Sloat, Amanda;
about it. It's about who we are. Closing comment: I was with
, he referred to Mayor Michael E. Duggan of Detroit, MI; President
; Obama, Barack; Walensky, Rochelle P.; Walke, Henry;
touch a smidgen of what the relationship with China really is about. I've known

the time I left office as Vice President, I had spent more time with
And the third thing, and the thing that I admire about dealing with with
Obama, Barack; Putin, Vladimir Vladimirovich; Trump, Donald J.;
— Q. Do you have a comment on the Georgia election law? President
/ Prime Minister Boris Johnson of the United Kingdom Q. Have you invited President
Washington, DC. Names: Johnson, Boris; Putin, Vladimir Vladimirovich;
ones. We just have to imagine again. I had a long discussion with
. Mitchell; Peduto, William M.; Wolf, Thomas W.;
you had a chance to speak to any of your international partners, any of—President
beginning? The President. No, I have not had that conversation with President
Anthony S. Fauci; Senate Minority Leader A. Mitchell McConnell; and President
, Anthony S.; Francis, Pope; McConnell, A. Mitchell;
. Secretary Blinken can tell you, I spent a lot of time with President

| Xi |
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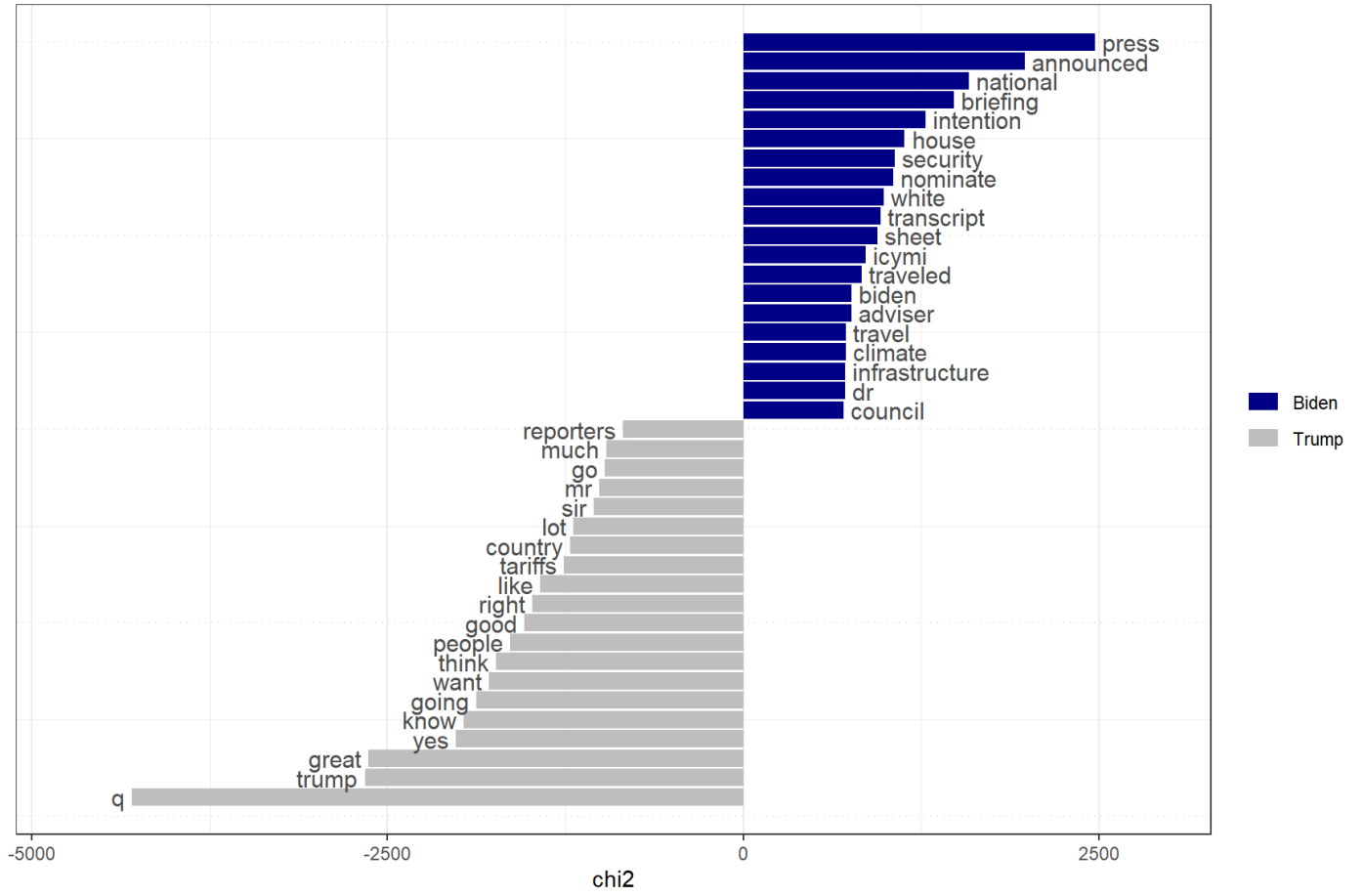
asked me on the Tibetan Plateau—I said," Yes, in one word:
Jinping of China; and White House Press Secretary Jennifer R. Psaki. He
Jinping. Subjects: China: President; Civil rights: Minorities:: Minority
Jinping in China. I spent more time with him, I'm told, than
Jinping of China, in his former capacity as Vice President of China; former
Jinping. Subjects: China: Prime Minister; Diseases: Coronavirus, domestic prevention
Jinping for a long time. Allegedly, by the time I left office as
Jinping than any world leader had, because President Obama and the Chinese President Hu
is, he understands—he makes no pretense about not understanding what I'm saying any more
Jinping. Subjects: Afghanistan: President; Afghanistan: U.S. military forces:
Jinping of China/ President Vladimir Vladimirovich Putin of Russia/ Prime Minister Boris Johnson
and Putin to the climate summit? The President. I haven't yet, but
Jinping. Subjects: China: President; Communications: News media:: Presidential
Jinping, the leader of China, when he called to congratulate me. We
Jinping. Subjects: Business and industry: Domestic investment, promotion efforts; China
, who I know you go way back with? Have you had the chance

```
. Thank you. NOTE: The President spoke at 4: 10 p.m.
Jinping of China. Categories: Addresses and Remarks: COVID–19 vaccination efforts; Interviews
Jinping; Yellen, Janet L. Subjects: China: President; Congress:
: traveled over 17,000 miles with him; spent, they tell me, over
```

Keyness by administration

```
[1] "Biden" "Trump"
```

	feature	chi2	p	n_target	n_reference
1	press	2475.0415	0	3214	350
2	announced	1981.4359	0	3019	487
3	national	1585.2195	0	3220	811
4	briefing	1481.4088	0	1633	87
5	intention	1281.8716	0	1253	15
6	house	1135.1080	0	3596	1368
7	security	1065.6664	0	2746	885
8	nominate	1054.6162	0	1016	7
9	white	989.9796	0	3123	1184
10	transcript	968.2509	0	1266	140
11	sheet	944.1099	0	1176	111
12	icymi	859.0758	0	851	14
13	traveled	832.8088	0	1330	234
14	biden	762.2290	0	2366	886
15	adviser	761.2694	0	1210	211
16	travel	722.5697	0	995	127
17	climate	720.4288	0	755	27
18	infrastructure	717.4210	0	893	84
19	dr	716.7668	0	813	51
20	council	704.8275	0	1014	144



2(a)(iii). Comparison of the two sources

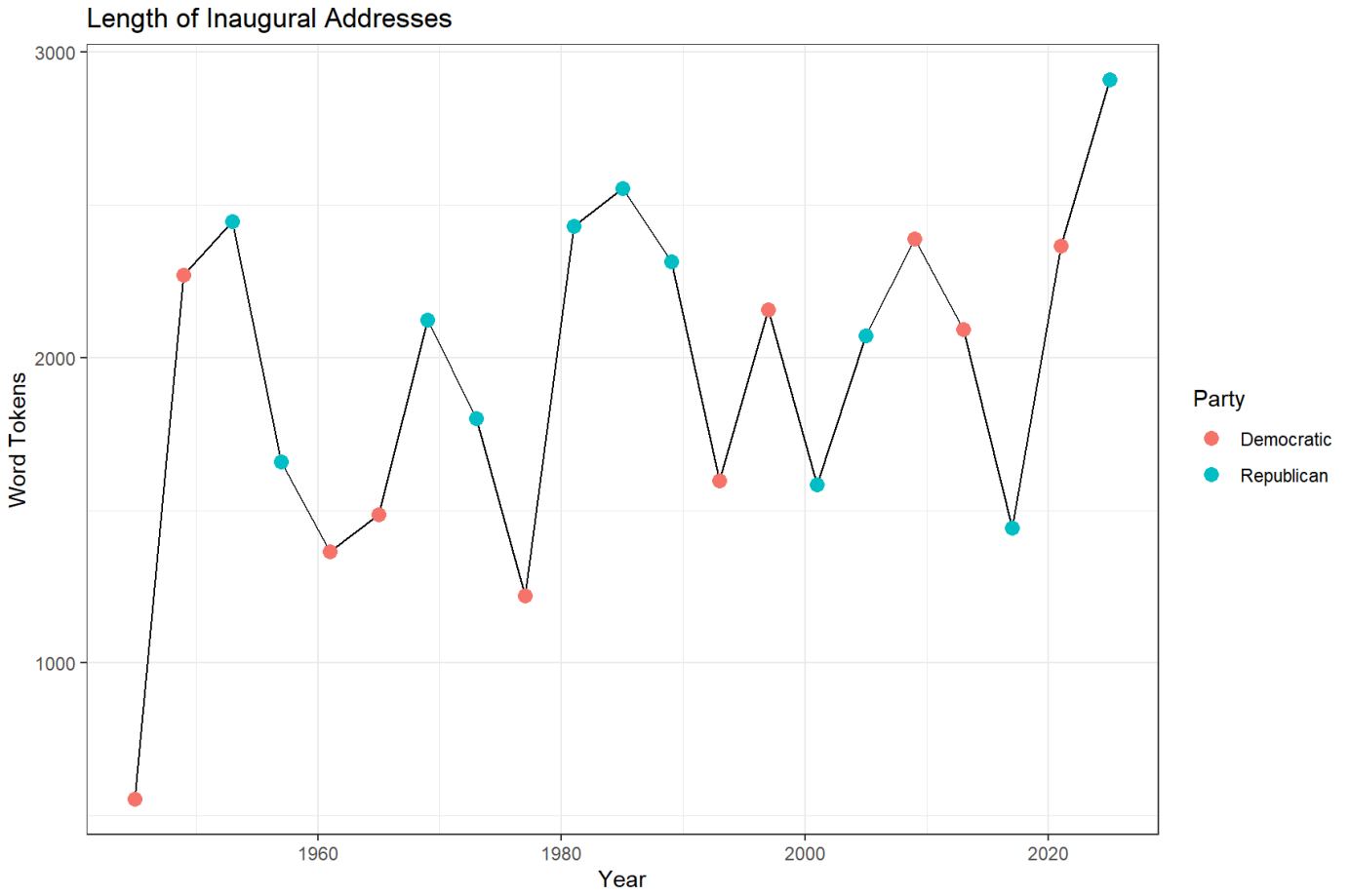
Interpretation will be added after reviewing the results.

2(b). Presidential inaugural addresses

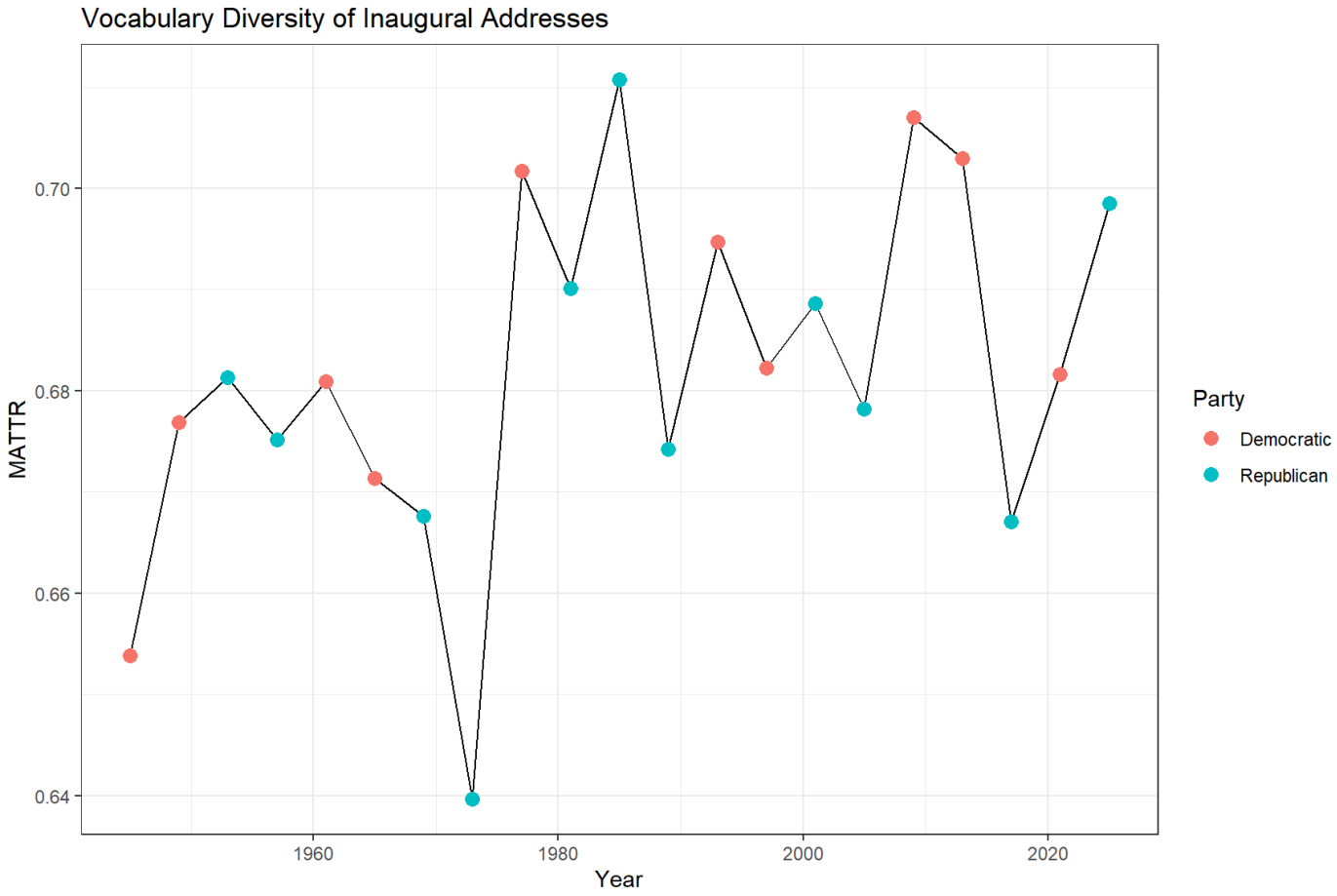
Speech length

document	Year	President	Party	words
1945-Roosevelt	1945	Roosevelt	Democratic	555
1949-Truman	1949	Truman	Democratic	2271
1953-Eisenhower	1953	Eisenhower	Republican	2445
1957-Eisenhower	1957	Eisenhower	Republican	1659
1961-Kennedy	1961	Kennedy	Democratic	1364
1965-Johnson	1965	Johnson	Democratic	1486
1969-Nixon	1969	Nixon	Republican	2124
1973-Nixon	1973	Nixon	Republican	1801

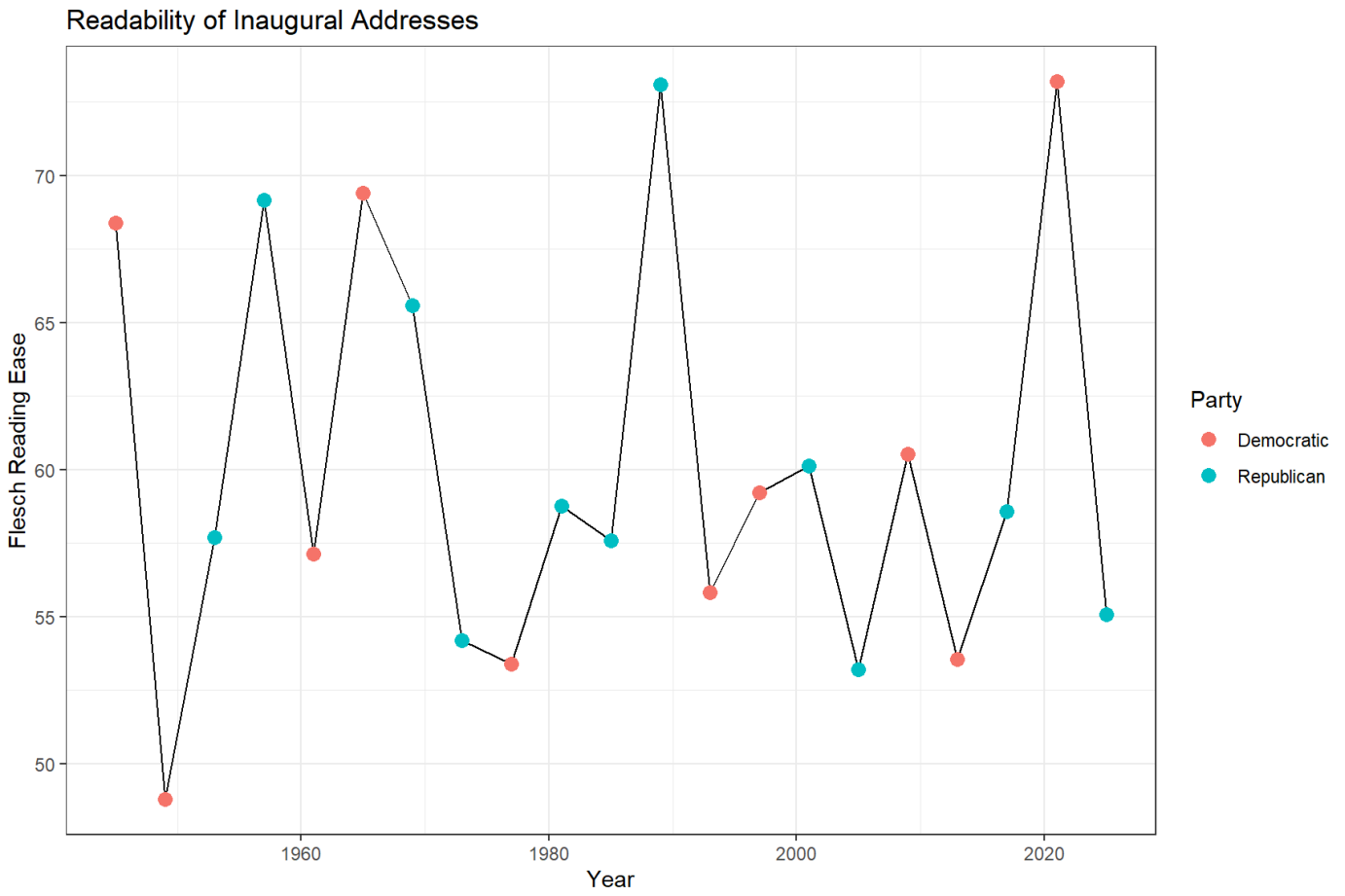
document	Year	President	Party	words
1977-Carter	1977	Carter	Democratic	1221
1981-Reagan	1981	Reagan	Republican	2431
1985-Reagan	1985	Reagan	Republican	2553
1989-Bush	1989	Bush	Republican	2315
1993-Clinton	1993	Clinton	Democratic	1598
1997-Clinton	1997	Clinton	Democratic	2157
2001-Bush	2001	Bush	Republican	1583
2005-Bush	2005	Bush	Republican	2071
2009-Obama	2009	Obama	Democratic	2390
2013-Obama	2013	Obama	Democratic	2092
2017-Trump	2017	Trump	Republican	1443
2021-Biden	2021	Biden	Democratic	2366
2025-Trump	2025	Trump	Republican	2909



Lexical diversity



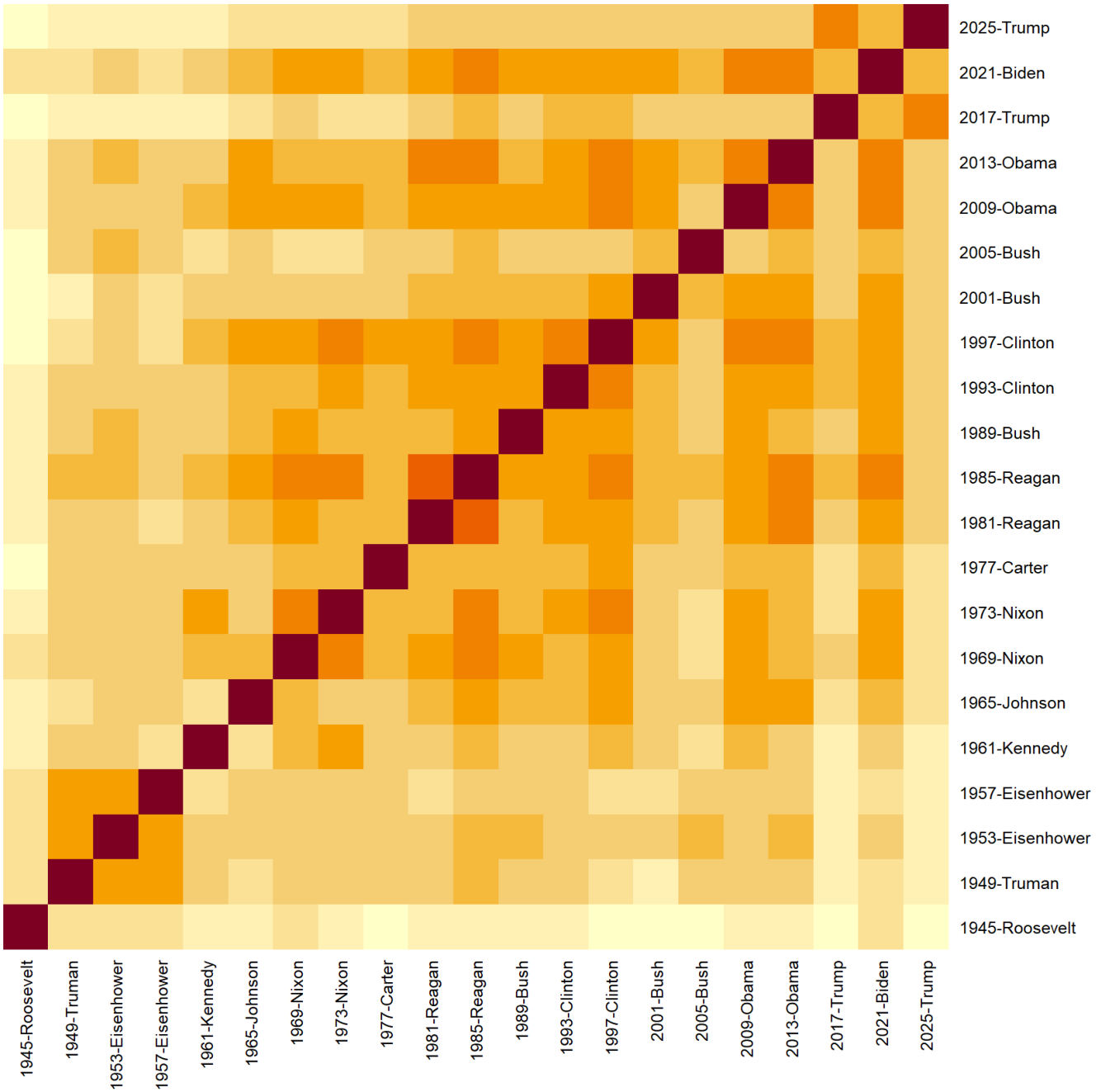
Readability



document	Year	President	Party	words	TTR	MATTR	Flesch
1945-Roosevelt	1945	Roosevelt	Democratic	555	0.463	0.654	68.380
1949-Truman	1949	Truman	Democratic	2271	0.326	0.677	48.794
1953-Eisenhower	1953	Eisenhower	Republican	2445	0.347	0.681	57.696
1957-Eisenhower	1957	Eisenhower	Republican	1659	0.353	0.675	69.164
1961-Kennedy	1961	Kennedy	Democratic	1364	0.391	0.681	57.120
1965-Johnson	1965	Johnson	Democratic	1486	0.354	0.671	69.408
1969-Nixon	1969	Nixon	Republican	2124	0.333	0.668	65.576
1973-Nixon	1973	Nixon	Republican	1801	0.282	0.640	54.188
1977-Carter	1977	Carter	Democratic	1221	0.401	0.702	53.382
1981-Reagan	1981	Reagan	Republican	2431	0.346	0.690	58.754
1985-Reagan	1985	Reagan	Republican	2553	0.336	0.711	57.584
1989-Bush	1989	Bush	Republican	2315	0.323	0.674	73.103
1993-Clinton	1993	Clinton	Democratic	1598	0.375	0.695	55.814
1997-Clinton	1997	Clinton	Democratic	2157	0.333	0.682	59.222
2001-Bush	2001	Bush	Republican	1583	0.369	0.689	60.120
2005-Bush	2005	Bush	Republican	2071	0.350	0.678	53.193

document	Year	President	Party	words	TTR	MATTR	Flesch
2009-Obama	2009	Obama	Democratic	2390	0.374	0.707	60.532
2013-Obama	2013	Obama	Democratic	2092	0.371	0.703	53.555
2017-Trump	2017	Trump	Republican	1443	0.373	0.667	58.576
2021-Biden	2021	Biden	Democratic	2366	0.308	0.682	73.196
2025-Trump	2025	Trump	Republican	2909	0.323	0.699	55.084

Cosine similarity

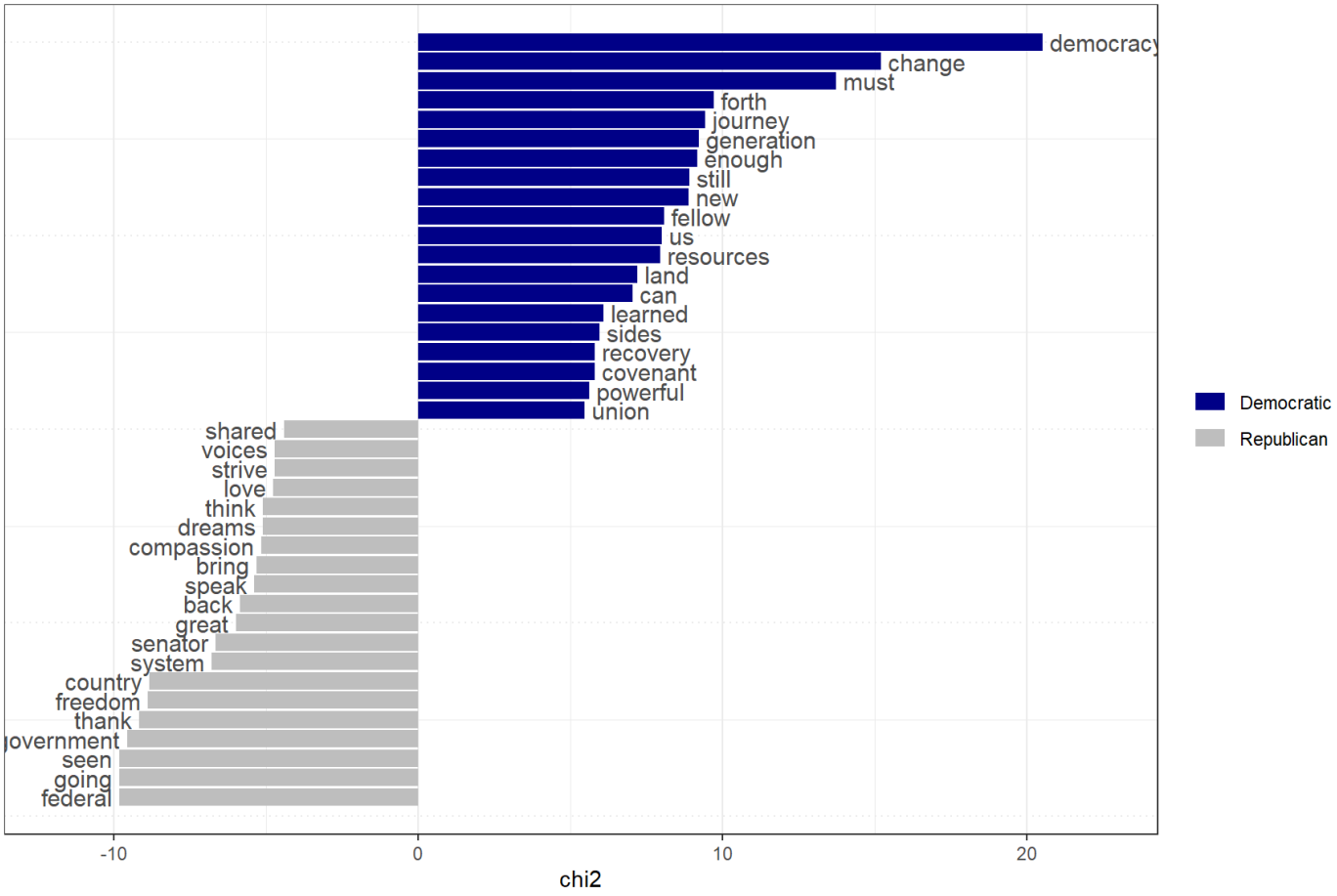


```
# A tibble: 10 × 3
   speech1      speech2      similarity
   <chr>        <chr>             <dbl>
 1 1981-Reagan  1985-Reagan       0.690
 2 1985-Reagan  1997-Clinton      0.642
 3 2009-Obama   2013-Obama        0.638
 4 1985-Reagan  2013-Obama        0.634
 5 2017-Trump   2025-Trump        0.632
 6 1969-Nixon   1973-Nixon        0.631
 7 1973-Nixon   1997-Clinton      0.630
 8 1997-Clinton 2009-Obama        0.630
 9 1993-Clinton 1997-Clinton      0.626
10 2009-Obama   2021-Biden        0.616
```

Keyness by party

```
[1] "Democratic" "Republican"
```

	feature	chi2	p	n_target	n_reference
1	democracy	20.516200	5.912871e-06	32	9
2	change	15.194647	9.697794e-05	30	11
3	must	13.715945	2.126415e-04	93	69
4	forth	9.697303	1.845387e-03	9	0
5	journey	9.412386	2.155246e-03	16	5
6	generation	9.204902	2.413679e-03	20	8
7	enough	9.163013	2.469556e-03	11	2
8	still	8.895073	2.859412e-03	21	9
9	new	8.885829	2.873926e-03	87	72
10	fellow	8.082754	4.468849e-03	32	19
11	us	7.989535	4.704849e-03	149	142
12	resources	7.943951	4.824839e-03	10	2
13	land	7.189682	7.332397e-03	24	13
14	can	7.032923	8.002461e-03	96	86
15	learned	6.084807	1.363493e-02	10	3
16	sides	5.957072	1.465836e-02	8	1
17	covenant	5.783959	1.617307e-02	6	0
18	recovery	5.783959	1.617307e-02	6	0
19	powerful	5.611930	1.783861e-02	11	4
20	union	5.462526	1.942855e-02	16	8



Latest address in the installed corpus

```
[1] '4.5.1'
```

```
[1] "2005-Bush"  "2009-Obama" "2013-Obama" "2017-Trump" "2021-Biden"
[6] "2025-Trump"
```

```
[1] 2025
```

```
[1] TRUE
```

3. Wordfish

Model preparation and orientation

```
[1] 4601
```

```
[1] 2093
```

```
[1] "1945-Roosevelt" "2025-Trump"
```

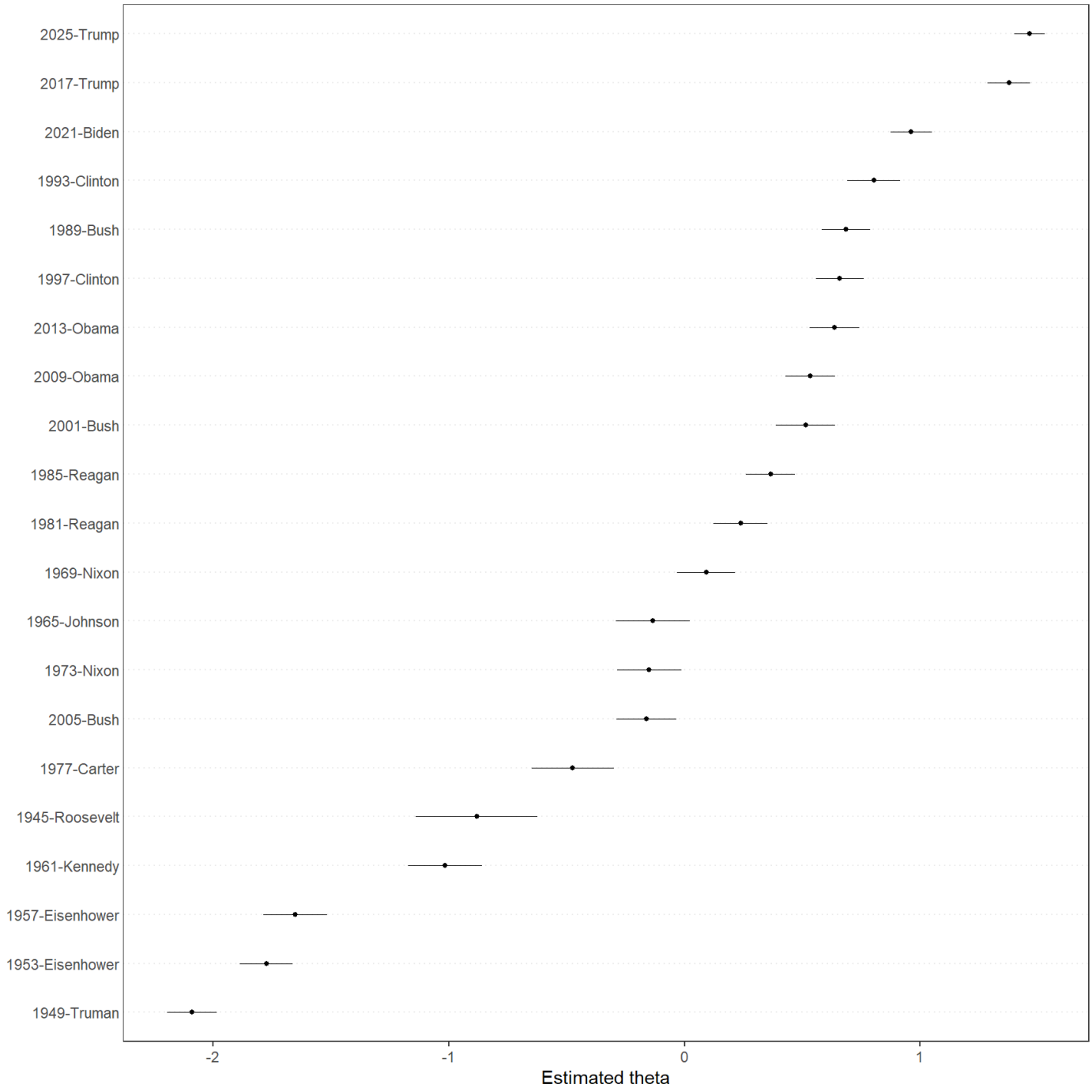
Model estimation

```
Call:
textmodel_wordfish.dfm(x = wordfish_dfm, dir = c(first_document,
    last_document))

Estimated Document Positions:
                   theta      se
1945-Roosevelt  -0.88100 0.13197
1949-Truman     -2.08959 0.05382
1953-Eisenhower -1.77381 0.05749
1957-Eisenhower -1.65098 0.06951
1961-Kennedy    -1.01553 0.08010
1965-Johnson    -0.13344 0.07993
1969-Nixon       0.09208 0.06283
1973-Nixon      -0.14966 0.06929
1977-Carter     -0.47387 0.08876
1981-Reagan      0.23760 0.05890
1985-Reagan      0.36501 0.05329
1989-Bush        0.68477 0.05246
1993-Clinton     0.80344 0.05736
1997-Clinton     0.65889 0.05138
2001-Bush        0.51455 0.06410
2005-Bush       -0.16030 0.06487
2009-Obama       0.53378 0.05406
2013-Obama       0.63602 0.05427
2017-Trump       1.37609 0.04611
2021-Biden       0.96165 0.04515
2025-Trump       1.46431 0.03284

Estimated Feature Scores:
       chief justice       mr    vice president friends understand believe
beta -0.1172 -0.4159 -0.56538 -0.1221    0.4513 -0.3122     0.3894 -0.4026
psi  -0.3884  0.7827  0.07029 -0.1003    0.8672  0.2005    -0.8349  0.6999
       agree    wish    form inauguration   simple  words americans  today
beta -0.5063 -0.8678 -0.1857      -0.1766 -0.01057 0.1405    0.4881 0.3228
psi  -1.5207 -1.5128 -1.1463      -1.1466 -0.47149 0.1873    1.1419 1.3375
     together  allies passing  period supreme   test courage  resolve  wisdom
beta   0.1362 -0.4988 -0.0145 -0.8503  -1.129 -1.101 0.13628 -0.17992 -0.6728
psi    1.1738 -1.0486 -1.7236 -1.3468  -1.914 -1.888 0.07519 -0.09669 -0.7941
     essential democracy    meet honorably   shall
beta    -1.027   -0.1893 -0.2637    -1.270 -0.8819
psi     -2.043    0.6215  0.2398    -2.286  0.6929
```

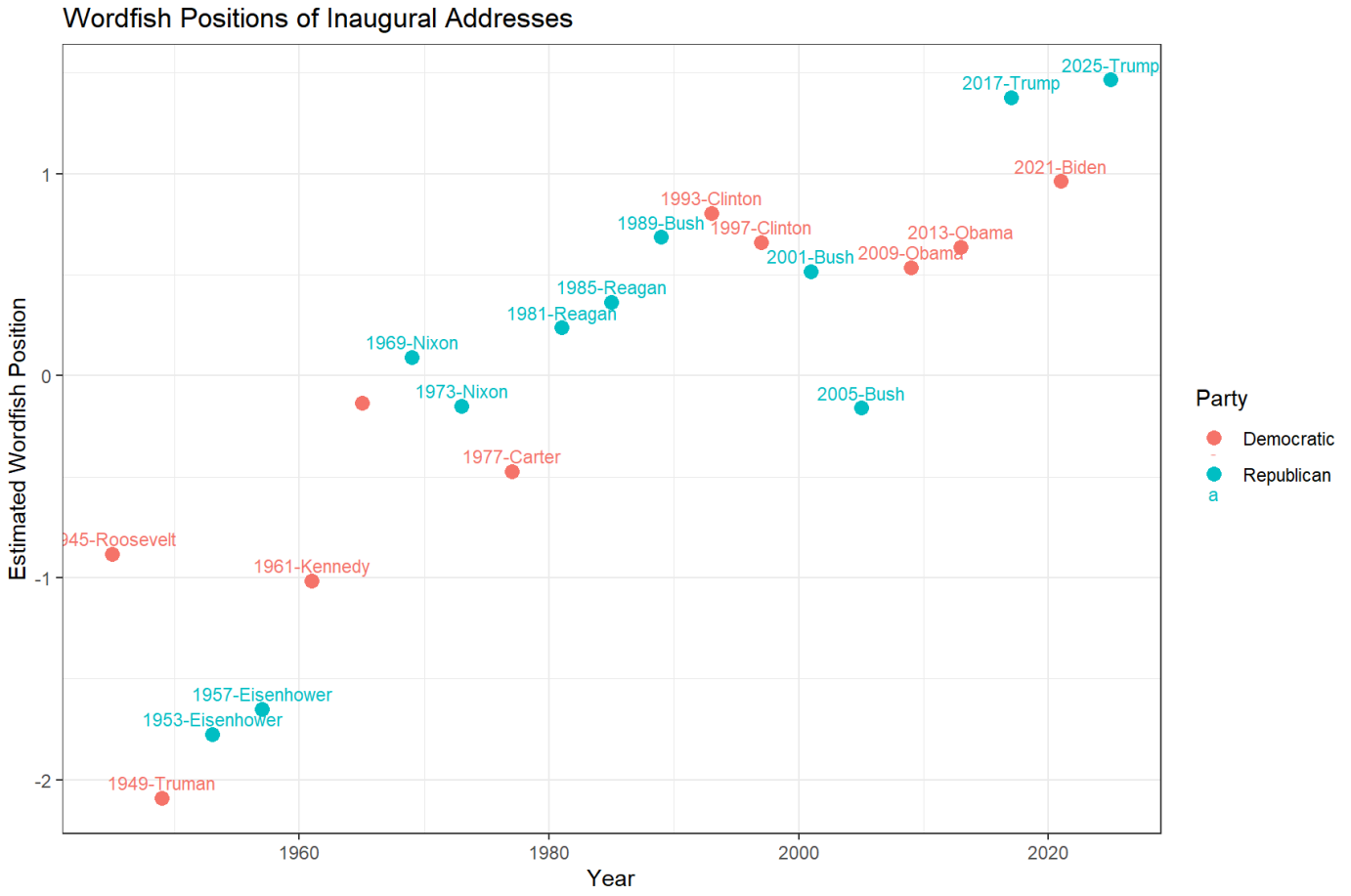
Estimated document positions



document	Year	President	Party	position	standard_error
1949-Truman	1949	Truman	Democratic	-2.090	0.054
1953-Eisenhower	1953	Eisenhower	Republican	-1.774	0.057
1957-Eisenhower	1957	Eisenhower	Republican	-1.651	0.070
1961-Kennedy	1961	Kennedy	Democratic	-1.016	0.080
1945-Roosevelt	1945	Roosevelt	Democratic	-0.881	0.132
1977-Carter	1977	Carter	Democratic	-0.474	0.089
2005-Bush	2005	Bush	Republican	-0.160	0.065
1973-Nixon	1973	Nixon	Republican	-0.150	0.069

document	Year	President	Party	position	standard_error
1965-Johnson	1965	Johnson	Democratic	-0.133	0.080
1969-Nixon	1969	Nixon	Republican	0.092	0.063
1981-Reagan	1981	Reagan	Republican	0.238	0.059
1985-Reagan	1985	Reagan	Republican	0.365	0.053
2001-Bush	2001	Bush	Republican	0.515	0.064
2009-Obama	2009	Obama	Democratic	0.534	0.054
2013-Obama	2013	Obama	Democratic	0.636	0.054
1997-Clinton	1997	Clinton	Democratic	0.659	0.051
1989-Bush	1989	Bush	Republican	0.685	0.052
1993-Clinton	1993	Clinton	Democratic	0.803	0.057
2021-Biden	2021	Biden	Democratic	0.962	0.045
2017-Trump	2017	Trump	Republican	1.376	0.046
2025-Trump	2025	Trump	Republican	1.464	0.033

Positions by year and party



Words associated with each end of the scale

```
# A tibble: 15 × 2
   word           beta
   <chr>         <dbl>
 1 basis         -2.97
 2 joint         -2.97
 3 conditions    -2.97
 4 europe        -2.57
 5 mockery       -2.49
 6 decrees       -2.49
 7 effective     -2.49
 8 participating -2.49
 9 aim           -2.49
10 materials     -2.49
11 undertaking   -2.49
12 imperialism   -2.49
13 concepts      -2.49
14 carrying      -2.49
15 firmness      -2.49

# A tibble: 15 × 2
   word         beta
   <chr>       <dbl>
 1 glorious     4.26
 2 obama        4.07
 3 radical      4.07
 4 disrepair    3.53
 5 ripped       3.53
 6 goodwill     3.53
 7 enforcement  3.53
 8 happen       3.28
 9 vote         3.22
10 deal         3.19
11 won          3.14
12 treated      3.14
13 roberts      3.04
14 rebuild      3.04
15 urban        3.04

[1] "1949-Truman"

[1] "2025-Trump"
```

▶ Read the two speeches at the extremes

CPD data note

Add source, permission, exact query, selection rules, and gaps after reviewing the collection records.

Collection log and manifest

The collection log and metadata manifest are the files to publish with this report. Do not publish the source PDFs, full-text corpus, or cache directory.

Collection log

Document manifest